B.Tech (1st Semester) **GROUP-C** **Batch (2026-30)**

Course Code-100103 **Engineering Chemistry** **3 0 0 3**

Course Outcomes (COs)	Course Outcome Statements
CO1	Predict the bonding and geometry of molecules using quantum-mechanical, MO and VSEPR principles.
CO2	Select suitable smart materials for engineering applications.
CO3	Apply polymer and rubber materials to solve engineering problems.
CO4	Apply adsorption and colloidal principles in chemical and industrial applications.
CO5	Apply green chemistry principles, water purification methods and adulteration testing methods in practical contexts.

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO1	*TSO 1a.* Explain quantization of energy and the experimental basis of the photoelectric effect. *TSO 1b.* Solve particle-in-a-box problems to determine energy levels. *TSO 1c.* Construct MO diagrams of a given molecule. *TSO 1d.* Predict the 3-D geometry and bond angle of a given molecule using VSEPR theory and Bent's Rule. *TSO 1e.* Determine coordination number, geometry, and possible isomers for given complexes.	**Unit- 1.0: Quantum Mechanics & Molecular Structure (9 hrs)** 1.1 Planck's theory, photoelectric effect, dual nature of electrons. 1.2 Wave Mechanics: Heisenberg Uncertainty Principle, Schrödinger's wave equation (concept & particle-in-a-box). 1.3 LCAO Method, MO diagrams of diatomics, VSEPR theory (geometries / Bent's Rule). 1.4 Coordination number, isomerism in transition-metal compounds; CFT (octahedral, tetrahedral and Square Planer).
CO2	*TSO 2a.* Explain the working principle of rotational, vibrational and electronic spectroscopy with an example. *TSO 2b.* Describe the role of UV-Vis, IR and NMR spectroscopy in identifying organic functional groups. *TSO 2c.* Interpret the UV-Vis, IR (fingerprint region) and ^{1}H-NMR spectrum of a given simple molecule.	**Unit- 2.0: Spectroscopy & Smart Materials Science (8 hrs)** 2.1 Spectroscopy: fundamentals of rotational, vibrational & electronic spectra; Beer–Lambert law. 2.2 Analytical Tools: UV-Vis, IR (fingerprint region) & NMR principles; applications in identifying organic functional groups. 2.3 Smart Materials: chemical sensors, piezoelectric materials used in MEMS & instrumentation

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
	TSO 2d. Explain the working principles of chemical sensors and piezoelectric materials. *TSO 2e.* Select an appropriate smart material (piezoelectric, chemical sensor) for a stated MEMS and instrumentation application.	
CO3	*TSO 3a.* Explain free-radical, ionic and condensation polymerisation mechanisms. *TSO 3b.* Classify polymers such as PVC, PS, PMMA and PTFE based on their properties and applications. *TSO 3c.* Explain the characteristics and applications of phenol-formaldehyde, epoxy resins and silicones. *TSO 3d.* Apply synthetic rubbers in suitable industrial and engineering applications.	**Unit- 3.0: Industrial Polymers & Advanced Materials (8 hrs)** 3.1 Introduction of Organic reactions, Polymerization: free-radical, ionic and condensation mechanisms; degree of polymerization. 3.2 Classification: thermoplastics vs. thermosets (PVC, PS, PMMA, PTFE). 3.3 Engineering Synthetics: Phenol-formaldehyde, Epoxy resins and Silicones. 3.4 Elastomers**:** synthetic rubbers (Buna-S, Buna-N, Neoprene).
CO4	*TSO 4a.* Explain the principles of electrochemical cells and electrode potentials. *TSO 4b.* Calculate the cell potential / EMF of a given electrochemical cell using the Nernst equation. *TSO 4c.* Explain the primary & secondary cells, Li-ion batteries, Pb-storage and Ni-Cd batteries. *TSO 4d.* Explain the working of an H_2-O_2 fuel cell, a solar cell, Lubricants (Flash point, Fire point, acid value. *TSO 4e.* Explain the mechanisms of corrosion and protective techniques. *TSO 4f.* Explain the properties and behaviour of colloidal systems and adsorption processes**.**	**Unit- 4.0: Electrochemistry, Corrosion & Colloidal (9 hrs)** 4.1 Electrochemical Theory: EMF, electrode potential, Nernst equation; numerical problems. 4.2 Storage Systems: primary & secondary cells, Li-ion batteries, Pb-storage and Ni-Cd batteries. 4.3 Next-Gen Energy: H_2-O_2 and methanol, (fuel cells, solar cells), Lubricants (Flash point, Fire point, acid value. 4.4 Material Durability: corrosion -waterline, stress, pitting, galvanic; protection, cathodic, anodic, coatings, e-coat. 4.5 Colloidal and adsorption
CO5	*TSO 5a.* Explain the principles of the EDTA method for hardness estimation and alkalinity determination.	**Unit- 5.0: Water Chemistry & Environmental Sustainability (8 hrs)** 5.1 Water Analysis: unit of concentration, estimation of hardness by EDTA, and alkalinity.

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	TSO 5b. Apply water purification and softening techniques in practical situations. *TSO 5c.* Explain the causes and effects of boiler scales, sludge, priming, foaming and caustic embrittlement. *TSO 5d.* Explain the impact of pollution, adulteration and e-waste on the environment and health.	5.2 Purification: lime-soda, zeolite & ion-exchange softening; measurement of BOD and COD. 5.3 Industrial Challenges: boiler scales, sludge, priming, foaming, caustic embrittlement. 5.4 Green Chemistry & Sustainability**:** principles, air/water pollution, adulteration (fat butter, sugar, turmeric powder and chilli powder), E-waste management.

Reference/Books

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
1.	Engineering Chemistry	Jain and Jain	Dhanpat Rai Publishing Co., 17th Ed. (ISBN: 978-9383182626)
2.	A Textbook of Engineering Chemistry	Shashi Chawla	Dhanpat Rai & Co., 2024 Ed. (ISBN: 978-8127262143)
3.	Principles of Materials Science and Engineering	William F. Smith	McGraw-Hill Education, 3rd Ed. (ISBN: 978-0070592025)
4.	Concise Inorganic Chemistry	J.D. Lee	Oxford University Press, 5th Ed. (ISBN: 978-0632052936)